Conditional expectations

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OLS

Ordinary Least Square, Geometric interpretation, It is very important, understand..

...

Think in A as an space transformation **operator**.

$$A\vec{x} = \vec{y}$$

The solution of the system depends on the "transformation" that apply A, what happens when $det(A) = 0$?

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Means that A Squishes the space (a lower dimension)!! (**At least one column is linearly dependent of others.**

Think the solution depend on determinant zero or different of zero.

...

Column space of the Matrix is the subspace generated by the columns of A. Namely $span(A)$.

Concepts

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The rank of matrix is the number of linear independent columns of the matrix.

full rank

Given the matriz $\underset{n\times p}{A}$ if the $rank(A) = p$ then the matrix is full rank.

null space

When a matrix is not full rank (the space is squished) then there is a set of vectors that land in $\vec{0}$ this set of vector are named the **null space** or **kernel** of the matrix.

...

the column rank is the dimension of column space.

...

In $\mathbb{R}^3$ any plane through the origin is a subspace.

...

the vectors $(1, 0)$ and $0, 1$ are a basis $\mathcal{B}$ for $\mathbb{R}^2$ but they are not, orthogonal the $\mathcal{B}'$ composed of $(1, 0)$ and $(0, 1)$ if is orthogonal.

The orthogonal complement to the subpace $W \in \mathbb{R}^n$ is defined as :

...

$$W^{\perp} = \{\vec{x} \mid \vec{x}.\vec{y} = 0, \forall \vec{y} \in W\} \tag{1}$$

complement is easier defined if you can test if a candidate vector $\vec{x}$ is orthogonal to W only proof that is orthogonal to each bases in $\mathcal{B}$.

$$\vec{x}.B_i = 0 \quad \forall i = 1, ..., k. \tag{2}$$

Remember that if we have a vector u and a subspace W we can write

...

$$u = project_W(u) + perp_w(u) \tag{3}$$

you can see graphically?

Remember that if we have a vector u and a subspace W we can write

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you can see graphically? Imagine in a 3D project a vector over a plane! in some texts also we can find that $perp_w(u) = w^{\perp}$.

...

if W is a subspace that belong to $\mathbb{R}^n$ and we want project u over it, then we need determine a **orthogonal basis** for W, the basis where $B_i \in \mathcal{B}$, such that

$$project_W(u) = \frac{B_1 u}{\|B_1\|^2} B_1 + ... + \frac{B_k u}{\|B_k\|^2} B_k \tag{4}$$

See the ilustration in the following slide.

...

note that

$$Perp_W(u) = u - project_W(u)$$

if by definition $project_W(u)$ belong to W, why $Perp_W(u)$ belong to $W^{\perp}$?

We can test:

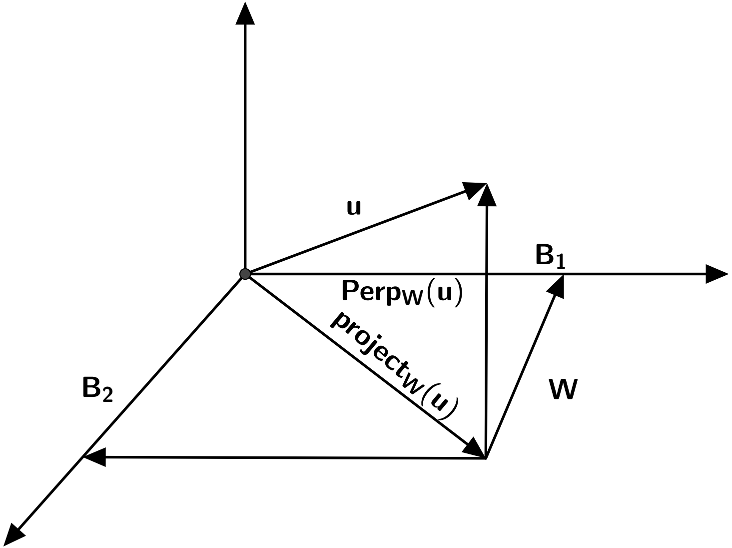
...

$$Perp_W(u).B_i = 0 \quad \forall i = 1, ..., k. \tag{5}$$

...

$$
\begin{aligned}
Perp(u).B_i &= 0 \\
(u - project_W(u))B_i &= uB_i - \frac{B_i u}{\|B_i\|^2} B_i.B_i \textbf{ (by orthogonal basis)} \\
&= 0
\end{aligned}
$$

Therefore is proof that $Perp_W(u) \in W^{\perp}$.


u
B_1
$Perp_W(u)$
$project_W(u)$
B_2
W

...

The previous picture we seen that the projected vector could be written as a linear combination of the basis vectors (of W)!. Note that $Perp_W$ is also perpendicular to any vector in W (including the basis).

The orthogonal complement to a subspace W is

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The null space of A;

...

$$N(A) = \{\vec{x} \in \mathbb{R}^n \mid A\vec{x} = 0\}$$

In words, any vector $\vec{x}$ in the null space is orthogonal to all rows A.

...

$$A\vec{x} = 0$$

$$\begin{pmatrix} r_1 \\ \ldots \\ r_n \end{pmatrix} \vec{x} = \begin{pmatrix} r1.\vec{x} \\ \ldots \\ r_n.\vec{x} \end{pmatrix} = \vec{0}$$

where r_i is the $i - th$ row of matrix A.

According to this

...

$$row(A)^{\perp} = N(A)$$
$$col(A)^{\perp} = N(A^T)$$

Review fundamental theorem of linear algebra.

Notation

if W and $\vec{x}$ are a subspace and a vector in $\mathbb{R}^n$respectively the projection of the vector $\vec{x}$ on W will be written as $\vec{x}_W$

...

$$\vec{x} = \vec{x}_W + \vec{x}_{W^\perp} \tag{6}$$

Given a matrix $\underset{m \times p}{A}$ and $W = col(A)$ being a $\vec{x} \in \mathbb{R}^m$, by definition $\vec{x}_W \in W = col(A)$ exist a vector $\vec{c}$ such that:

$$A\vec{c} = \vec{x}_w \tag{7}$$

By definition $\vec{x} - A\vec{c} \in W^{\perp}$ the orthogonal compliment of W is equal to $Null(A^T)$

...

$$A^T(\vec{x} - A\vec{c}) = 0$$
$$A^T\vec{x} - A^TA\vec{c} = 0$$

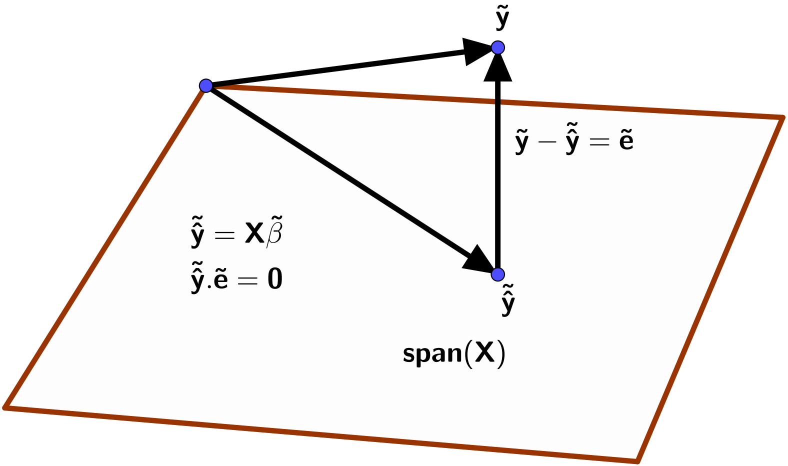
then we have that $\vec{x}_W = A(A^TA)^{-1}A^T\vec{x}$.

In the previous slide we derived $\vec{x}_W$ the projection of $\vec{x}$ on the subspace W, where $W = col(A) = span(A)$. Note that:

...

$$\hat{y} = X(X^T X)^{-1} X^T y \tag{8}$$

$\hat{y}$ is the orthogonal projection of y onto the subspace generated by $col(X)$.


$\tilde{\mathbf{y}}$
$\tilde{\mathbf{y}} - \tilde{\hat{\mathbf{y}}} = \tilde{\mathbf{e}}$
$\tilde{\hat{\mathbf{y}}} = \mathbf{X}\tilde{\beta}$
$\tilde{\hat{\mathbf{y}}}.\tilde{\mathbf{e}} = \mathbf{0}$
$\tilde{\hat{\mathbf{y}}}$
$\mathbf{span}(\mathbf{X})$

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In this lecture we are interested in